

Supplementary material

Apparent sequence-model contribution depends strongly on negative-set construction: a calibration across 94 ENCODE eCLIP datasets

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Overview

This document contains ten supplementary figures and one supplementary table. Each legend defines the panels, sample size, uncertainty and source table so that it can be read independently of the main text.

Two conventions hold throughout and are not repeated in every legend.

The estimand. The *nested contribution* of a model under a protocol is the pooled out-of-fold AUROC of a logistic model on 19 composition features plus that model’s score, minus the AUROC of the 19 features alone. Composition is mononucleotide and dinucleotide frequencies plus Shannon entropy, with one level dropped from each frequency family because frequencies sum to one and keeping every level makes the design matrix singular: $3 + 15 + 1 = 19$ columns. Higher means the sequence model adds more over knowing base composition.

Uncertainty. Unless a legend says otherwise, intervals are 95% percentile intervals from a bootstrap that resamples *proteins* and takes all datasets of each sampled protein, 4000 draws. The panel is 94 datasets over 79 proteins, 15 of which appear in both cell lines, so resampling datasets would treat correlated rows as independent evidence.

Estimator. Figures showing a span or contribution use the two-stage estimator unless stated otherwise, consistent with common practice in the literature. The paper’s primary estimator is the cross-fitted one; the two differ by about a ninth of the span and the main text reports both.

Figure and table numbering

The repository stores figures under their build names. The mapping below links the supplementary numbering to those filenames and is checked by `tests/unit/test_supplement.py`.

supplementary	repository file (in <code>results/figures/</code>)	subject	section
Figure S1	<code>f0_panel_overview</code>	the panel and how it was drawn	Methods
Figure S2	<code>f1_cost_of_matching</code>	the two-protocol contrast	3.1
Figure S3	<code>f2_four_models</code>	four model classes, one protocol	Methods
Figure S4	<code>f3_strand_placebo</code>	the strand control	Limitations
Figure S5	<code>f4_replication</code>	cross-cell-line replication	3.1
Figure S6	<code>f5_size_modification</code>	effect modification by size	3.1
Figure S7	<code>f6_k_sweep</code>	sensitivity to k	Methods
Figure S8	<code>f7_region</code>	region heterogeneity	3.1
Figure S9	<code>f8_scale_check</code>	the compression correction	3.2
Figure S10	<code>f13_baseline_order_models</code>	order-3 baseline by model class	3.3
Table S1	<code>supplementary_table_s1.csv</code>	dataset to ENCODE accession	Data avail.

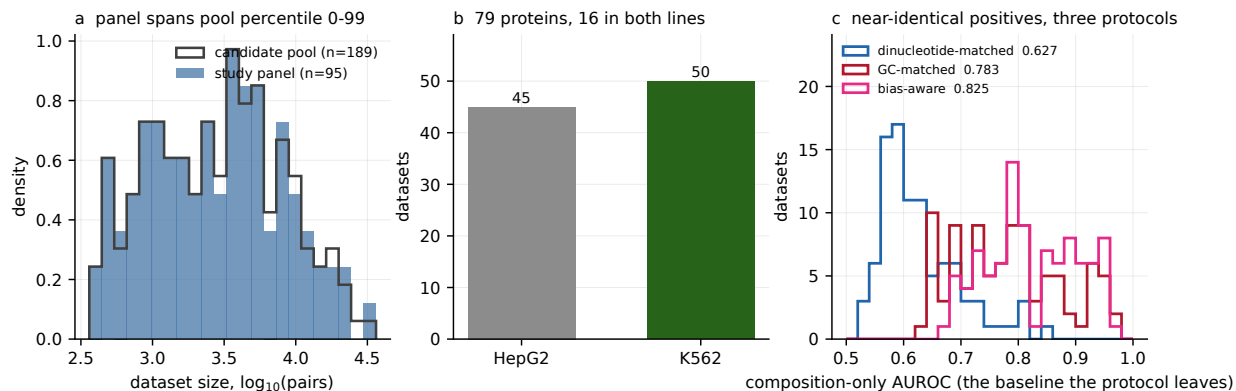


Figure S1: Study panel and size distribution. (a) Size distribution of the 94 study datasets, as log₁₀ matched pairs, against the candidate pool they were drawn from, shown as a step outline. The panel was selected as a deterministic systematic subset by pair rank, which guarantees coverage of the size range; it is not a probability sample and cannot be assumed free of aliasing against structure ordered by pair count. (b) Composition of the panel by protein and cell line. 79 distinct proteins, of which 15 are assayed in both K562 and HepG2 and therefore contribute two datasets each. This is why every interval in the study resamples proteins rather than datasets. (c) Composition-only AUROC under each of the three protocols, one point per dataset. This is a measured property of each protocol: the baseline each protocol leaves for a sequence model to improve on. **n:** 94 datasets, 79 proteins, 2 cell lines. **Uncertainty:** none plotted; panels show distributions rather than estimates. **Source:** `panel_summary.csv`, `candidate_sizes.csv`, `three_arm_per_dataset.csv`.

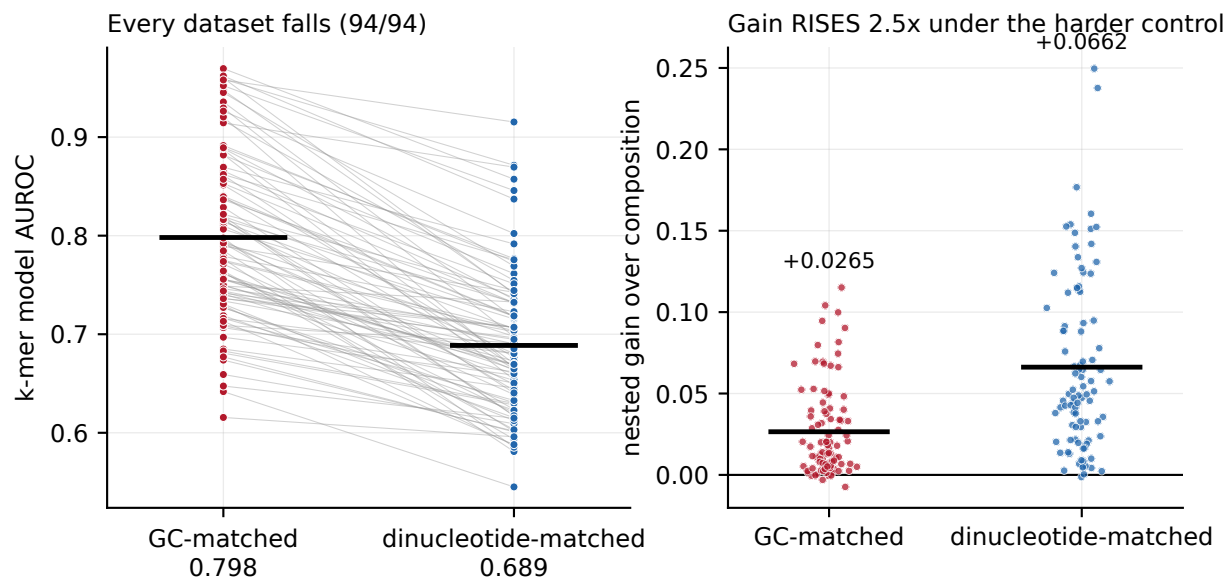


Figure S2: Effect of GC-matched versus dinucleotide-matched negatives on apparent AUROC and measured contribution. Paired within dataset, because the two arms share nearly all of their positives; an unpaired plot would discard the structure that makes the per-dataset counts meaningful. **(a)** The 4-mer model's own AUROC under each protocol. It falls in every dataset, which is expected: dinucleotide matching removes the compositional cue the model was partly using. **(b)** The nested contribution over composition under each protocol. It rises. The model is refitted under each protocol, and the positive sets are nearly but not perfectly identical because each matcher rejects positives it cannot pair. Their median Jaccard similarity is 0.9972, with exact agreement in 10 of 94 datasets. **n:** 94 paired datasets. **Uncertainty:** not plotted; the panel-level intervals are in the main text. **Source:** `cost_of_matching.csv`.

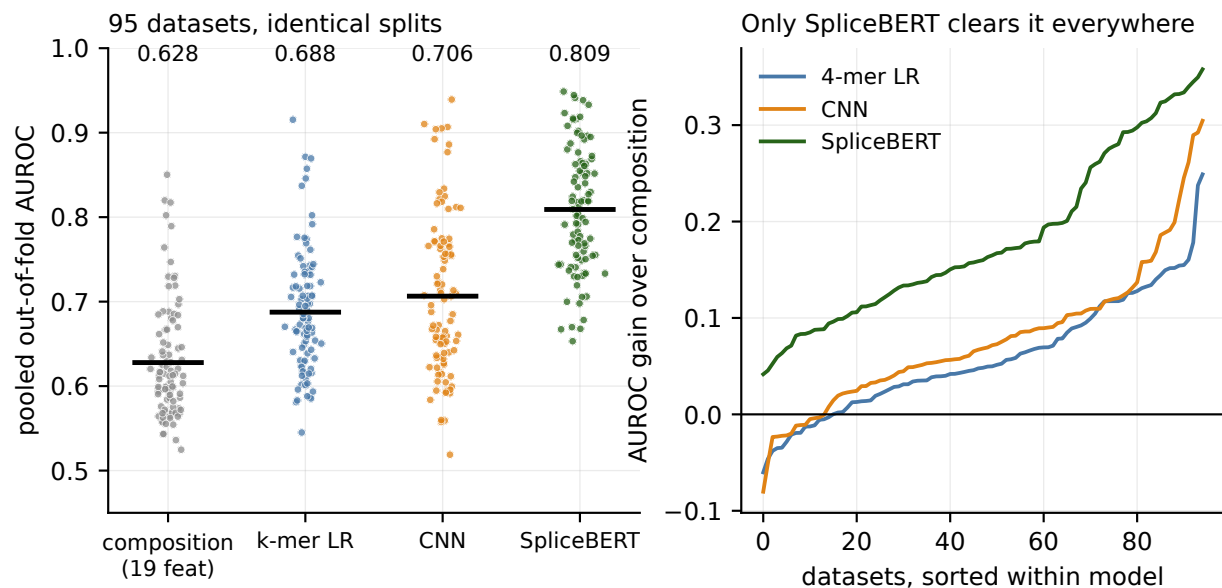


Figure S3: Four model classes under one protocol, on identical rows and folds. (a) Pooled out-of-fold AUROC for the 19-feature composition baseline, the 4-mer logistic model, the convolutional network and fine-tuned SpliceBERT. Within an arm all four are scored on the same rows under the same chromosome-grouped partition. (b) The same four as a gain over the composition baseline, with datasets sorted within each model. Only SpliceBERT exceeds the baseline in every dataset. This methods-oriented figure establishes the performance ordering under one protocol before the three-protocol comparison in the main text. **n**: 94 datasets. **Uncertainty**: not plotted; each point is one dataset's pooled out-of-fold AUROC. **Source**: `matched_four_models.csv`.

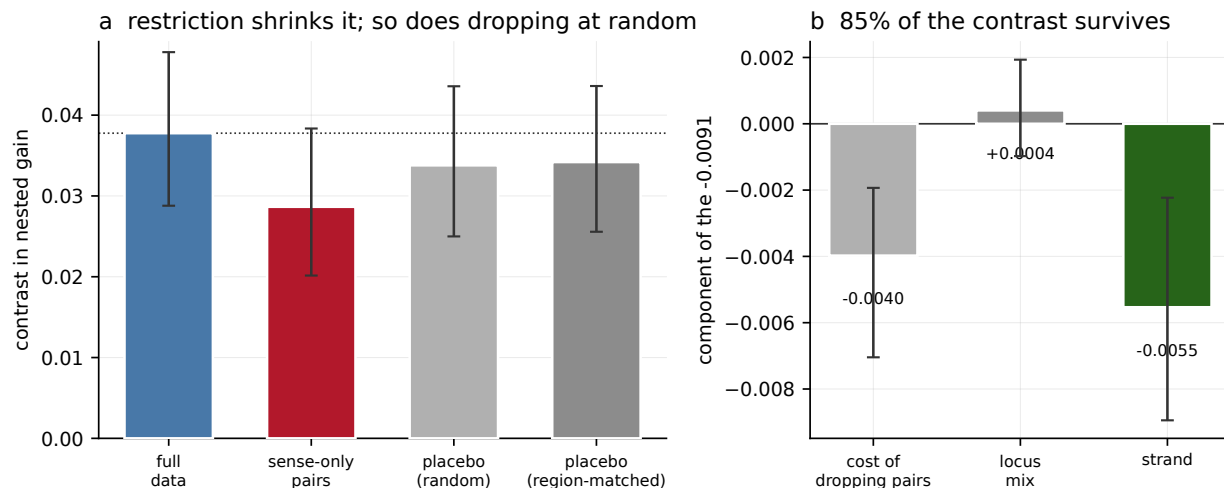


Figure S4: Strand-control analysis. Negatives inherit the strand of the positive they are matched to rather than the strand of the gene they sit in, so a fraction of them are antisense. Restricting to pairs whose negative is unambiguously on its own gene's strand discards most of the data, and a 256-feature model loses more from that restriction than a 19-feature baseline does, in both arms. **(a)** The contrast under four conditions: the full data, the sense-only restriction, a placebo dropping the same number of pairs at random, and a placebo matched on transcript region crossed with gene-density quartile. The naive control attributes the full reduction to strand, whereas the two placebos quantify the effect of data removal. **(b)** The same information as a decomposition of the -0.0091 that restriction alone reports, into the cost of dropping pairs at all and the strand-specific remainder. Gene density is included because retention requires exactly one overlapping gene strand and therefore selects against multi-gene loci by construction; the region-only placebo does not remove that selection. The strand-specific excess is -0.0055 , and 85.3% of the contrast remains after correction. **n:** 40 datasets, over 40 distinct proteins, so protein clustering is a no-op here and is asserted rather than assumed. 20 placebo seeds per dataset per arm. **Uncertainty:** 95% percentile intervals over datasets. **Source:** `strand.placebo.csv`.

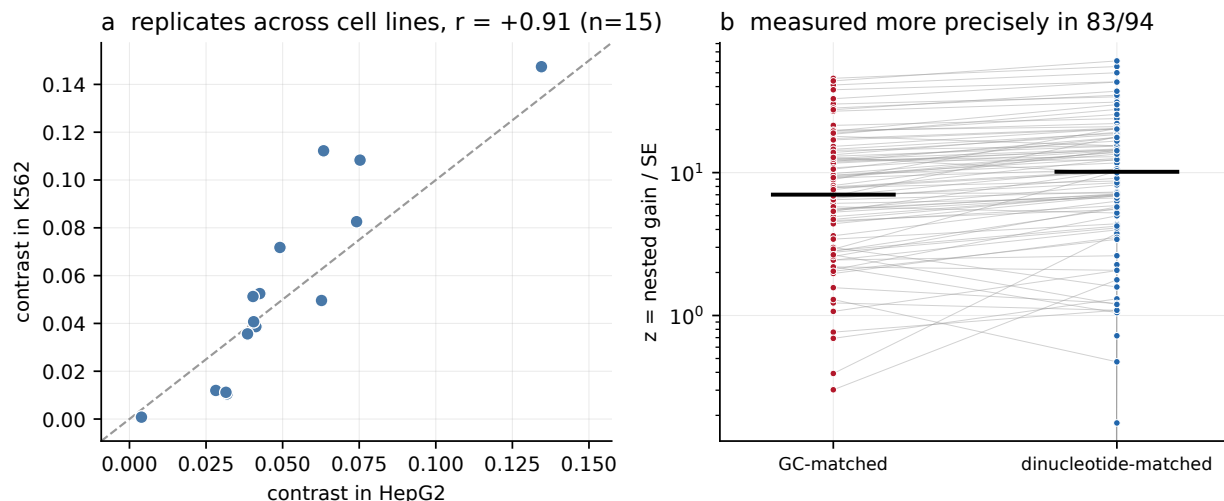


Figure S5: The contrast replicates across cell lines, and is measured more precisely under the harder protocol. (a) For the 15 proteins assayed in both K562 and HepG2, the contrast measured in one cell line against the contrast measured in the other. These are separate eCLIP experiments with separately drawn negatives, so one line is an out-of-sample prediction of the other. Pearson $r = +0.9094$. The design forces the sign in each cell line independently and guarantees nothing about whether the magnitudes agree, which is what makes this evidence rather than arithmetic. **(b)** $z = \text{gain}/\text{SE}$ under each protocol, paired per dataset. The gain rises faster than its standard error, so the same contribution reaches the same confidence with fewer labelled windows. After partialling out each dataset's total nested gain, the correlation is $+0.3321$ with $p = 0.2265$, so the replication is not independent of overall signal strength. **n:** 15 proteins for (a), 94 datasets for (b). **Uncertainty:** Pearson correlation with a protein-level bootstrap interval in the main text; DeLong standard errors for z . **Source:** `r1_robustness.csv`, `cost_of_matching.csv`.

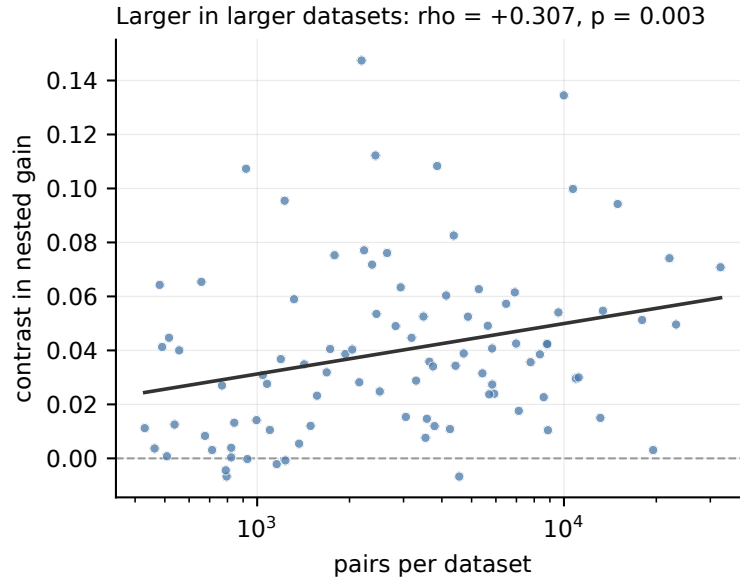


Figure S6: The contrast is larger in larger datasets. Each point is one dataset: matched pairs on a logarithmic axis against the contrast in nested contribution between the dinucleotide-matched and GC-matched protocols. The contrast is paired within dataset, and the horizontal axis uses the GC arm's pair count. Arm sizes differ slightly because the two matchers retain different positives; dataset size primarily affects precision rather than bias. It is reported here as effect modification, which is a different and weaker statement: the protocol matters more where there is more data. **n:** 94 datasets. **Uncertainty:** Spearman correlation, printed in the panel title; the protein-clustered interval is in the main text. **Source:** `cost_of_matching.csv`, `robustness.csv`.

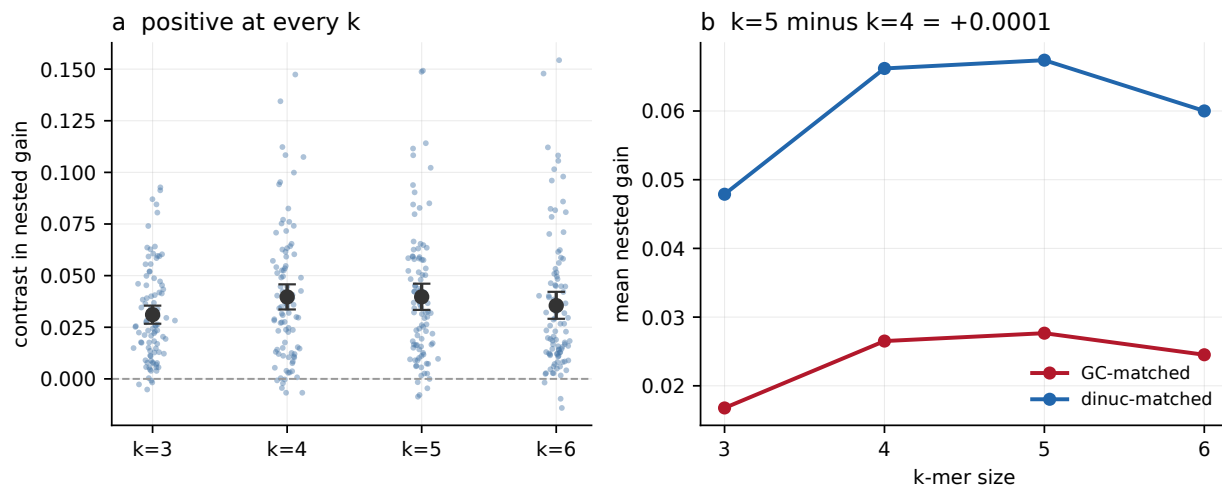


Figure S7: The contrast does not depend on the choice of k . The composition baseline and the k -mer model are refitted from raw sequence at four orders for all 94 datasets in both arms, rather than read from the tables the main analysis wrote. Together with `recompute.py` this is one of only two results in the study that are re-derived along a different path rather than compared against a record of themselves; the rebuild reproduces the published contrast to 1.2×10^{-6} . **(a)** The contrast at $k = 3, 4, 5, 6$, one point per dataset. Positive at every order, and positive at every order in 82 of 94 datasets individually. **(b)** The mean nested gain in each arm as a function of k , so a reader can see where the contrast comes from rather than only its difference. The difference between the $k = 4$ and $k = 5$ specifications is $+0.0001$ (paired Wilcoxon $p = 0.84$). The contrast is also smallest at $k = 3$, so order 4 is not the most favourable choice available. **n:** 94 datasets per order, both arms. **Uncertainty:** 95% protein-clustered percentile intervals in the source table. **Source:** `k_sweep.csv`, `k_sweep_per_dataset.csv`.

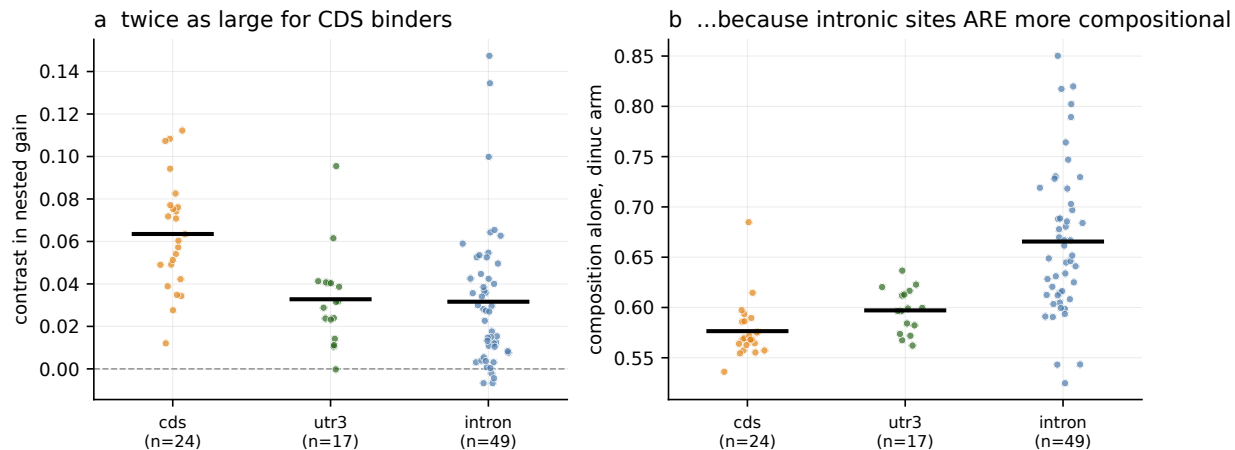


Figure S8: The contrast is larger for datasets whose binding sites are mostly coding, and the mechanism is checkable in the same figure. Datasets are grouped by the transcript region their positives mostly occupy. **(a)** The contrast by dominant region. CDS-dominant datasets give +0.0635 against +0.0316 for intron-dominant ones, a difference of +0.0319, which is as large as the headline contrast itself. **(b)** Intronic sites are compositionally distinctive, so composition alone already discriminates them and leaves the protocol less to expose. Composition-only AUROC is 0.6656 for intron-dominant datasets against 0.5765 for CDS-dominant ones. The intronic fraction correlates with the contrast at -0.3399 , but partialling out total nested gain removes the association. Region indexes how much non-compositional signal exists; it is not an independent mechanism. **n:** 94 datasets, grouped into three region classes with at least 8 datasets each. **Uncertainty:** Mann-Whitney and Kruskal-Wallis tests in the source table. **Source:** `region_heterogeneity.csv`, `region_heterogeneity_per_dataset.csv`.

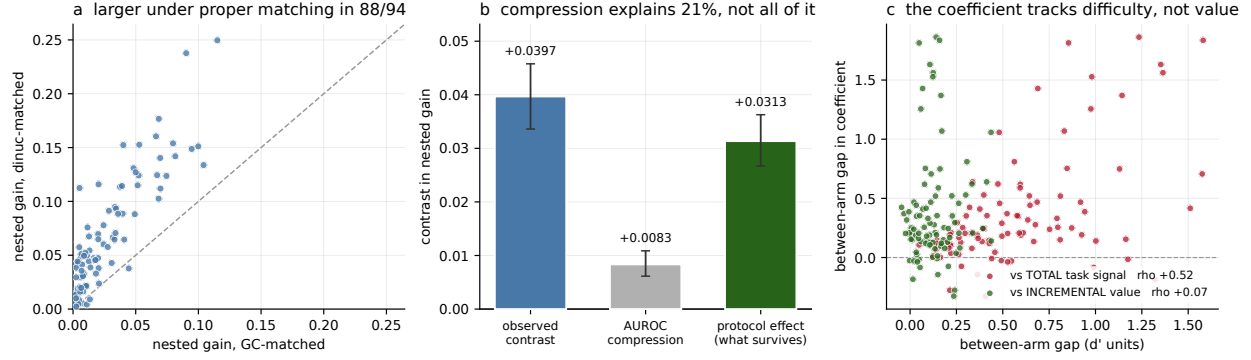


Figure S9: Sensitivity of the protocol contrast to baseline compression. An AUROC increment is bounded above by the headroom the baseline leaves, so part of any comparison between arms with unequal baselines is arithmetic. The transplant converts AUROC to d' , moves one arm's increment onto the other arm's baseline, and asks what remains. **(a)** The contrast per dataset, which needs no transplant and no link function: it is a difference of two AUROC differences on the same rows. **(b)** The transplant. Compression is real, a factor of 1.5081, and does not account for the contrast: the protocol effect net of scale is +0.0313 in one direction. **(c)** The diagnostic behind the log-odds reversal. The between-arm coefficient gap tracks each task's total signal rather than the incremental value it should measure, which is the standard non-comparability of logistic coefficients across separate fits. Running the transplant in the other direction gives +0.0215, and across six monotone links and both directions the smallest member is +0.0188. Every member keeps the sign, so the contrast is not a compression artefact. The range of estimates supports treating the decomposition as a sensitivity analysis. Separately, the identifying assumption that the d' increment is baseline-invariant is refuted by its own diagnostic, so no model's protocol effect is point-identified. **n:** 94 datasets. **Uncertainty:** 95% protein-clustered percentile intervals. **Source:** `scale_check.csv`, `protocol_identification.csv`.

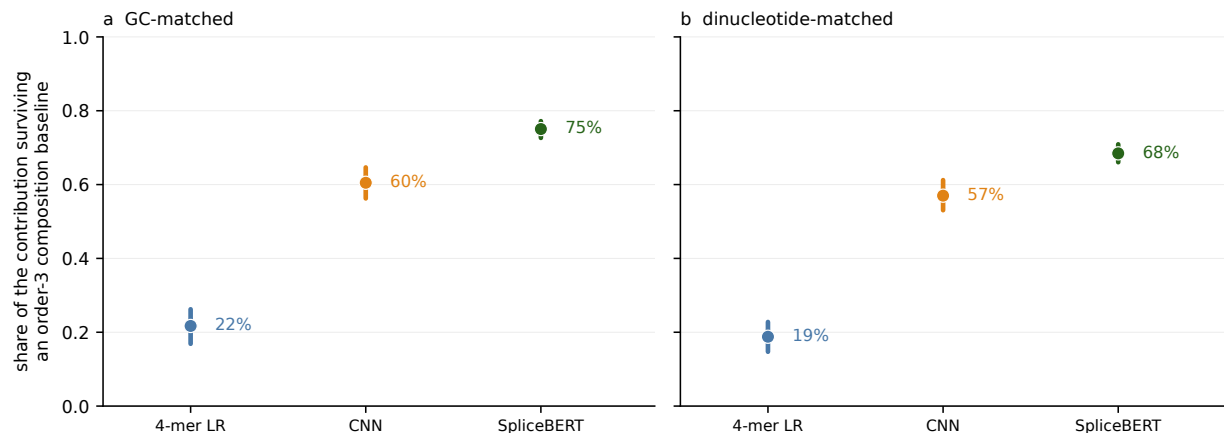


Figure S10: Raising the composition baseline from order 2 to order 3, for all three model classes. The composition baseline stops at dinucleotides while the headline model is a 4-mer, so the measured quantity is short-range compositional signal one order above wherever the matcher stopped. Where the baseline stops is an analyst's choice, and this figure measures what that choice costs. The extended baseline adds 63 non-redundant trinucleotide-frequency columns, for 82 features in total. **(a)** GC-matched arm and **(b)** dinucleotide-matched arm: the share of each model's contribution that survives an order-3 baseline. For the 4-mer the surviving share on the GC arm is 0.2170, against 0.6049 for the convolutional network and 0.7504 for SpliceBERT. The *absolute* amount absorbed is statistically indistinguishable across model classes, and the convolutional network loses the least of the three, which no capacity story predicts. The shares follow from the totals. What the figure supports is that the protocol contrast survives an order-3 baseline for every model class, not that the 4-mer is uniquely fragile. **n:** 94 datasets, three model classes, two arms. **Uncertainty:** 95% protein-clustered percentile intervals in the source table. **Source:** `baseline_order_models.csv`, `baseline_order_models_per_dataset.csv`.

Table S1. Dataset to ENCODE accession

The table below is typeset from `supplementary_table_s1.csv`, which ships beside this document and is committed under `results/tables/`. The CSV carries six further columns, each arm's composition baseline and contribution, which do not fit a portrait page; the accession mapping is reproduced here so that this document is readable on its own.

One row has the three-arm flag false: NCBP2 in K562 cleared the minimum-pair floor under dinucleotide matching and not under GC matching, so it is in the candidate panel and has no three-protocol result. That is the difference between the 95 selected datasets and the 94 the study reports on.

Table S1: Dataset to ENCODE accession. All 95 datasets in the candidate panel, one row each, with the ENCODE file and experiment accessions needed to retrieve the source peaks. **3 arms** is whether the dataset carries all three negative-set protocols; the 1 marked *no* cleared the minimum-pair floor under dinucleotide matching but not under GC matching, which is the difference between the 95 selected datasets and the 94 the study reports on. The same table with each arm's composition baseline and contribution is `supplementary_table_s1.csv`, which ships beside this document.

protein	cell line	ENCFF file	ENCSR experiment	reps	pairs	3 arms
AATF	K562	ENCFF955LYD	ENCSR819XBT	2	1,081	yes
AKAP1	HepG2	ENCFF749IUB	ENCSR356ZMO	2	5,943	yes
AQR	HepG2	ENCFF323PUC	ENCSR018WPY	2	11,101	yes
BCCIP	HepG2	ENCFF134LIU	ENCSR485QCG	2	539	yes
BCLAF1	HepG2	ENCFF810EUA	ENCSR876EYA	2	32,387	yes
BUD13	K562	ENCFF262ILV	ENCSR663WES	2	8,850	yes
CDC40	HepG2	ENCFF027MEO	ENCSR815VVI	2	1,172	yes
CSTF2T	HepG2	ENCFF141DOI	ENCSR919HSE	2	18,864	yes
DDX24	K562	ENCFF537KWB	ENCSR999WKT	2	9,600	yes
DDX55	HepG2	ENCFF797QRS	ENCSR845VGB	2	1,686	yes
DDX55	K562	ENCFF462WAI	ENCSR923NKN	2	1,101	yes
DDX59	HepG2	ENCFF899AWZ	ENCSR214BZA	2	3,089	yes
DDX6	HepG2	ENCFF614NUP	ENCSR141OIM	2	493	yes
DDX6	K562	ENCFF904KTV	ENCSR893EFU	2	1,955	yes
DGCR8	K562	ENCFF023YVD	ENCSR947JVR	2	1,253	yes
DHX30	K562	ENCFF128AKC	ENCSR529GSJ	2	824	yes
DROSHA	HepG2	ENCFF854BHP	ENCSR834YLD	2	2,184	yes
DROSHA	K562	ENCFF797BHU	ENCSR653HQC	2	3,915	yes
EFTUD2	HepG2	ENCFF796RMF	ENCSR527DXF	2	7,401	yes
EIF3H	HepG2	ENCFF813JIK	ENCSR916XIV	2	2,832	yes
ELAC2	K562	ENCFF850ZSC	ENCSR440MSY	2	12,915	yes
ELAVL1	K562	ENCFF566LNK	ENCSR090LNQ	2	8,806	yes
EXOSC5	HepG2	ENCFF862EKD	ENCSR693JWP	2	727	yes
FASTKD2	HepG2	ENCFF498IGK	ENCSR023UHL	2	868	yes
FASTKD2	K562	ENCFF423SOR	ENCSR887FHF	2	542	yes
FKBP4	HepG2	ENCFF957AOZ	ENCSR018ZUE	2	1,454	yes
FMR1	K562	ENCFF630KXX	ENCSR554UVB	2	4,445	yes
FXR1	K562	ENCFF366UJP	ENCSR774RFN	2	1,425	yes

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Table S1, continued

protein	cell line	ENCFF file	ENCSR experiment	reps	pairs	3 arms
FXR2	K562	ENCFF111LMZ	ENCSR767OEY	2	4,745	yes
G3BP1	HepG2	ENCFF899HHP	ENCSR721HPX	2	6,482	yes
GARS	K562	ENCFF390NMJ	ENCSR448GXJ	2	566	yes
GEMIN5	K562	ENCFF050VVI	ENCSR238CLX	2	2,640	yes
GRSF1	HepG2	ENCFF929AWR	ENCSR668MJX	2	1,008	yes
GRWD1	HepG2	ENCFF159MMZ	ENCSR893NWB	2	22,078	yes
GRWD1	K562	ENCFF327DLL	ENCSR999YGP	2	4,375	yes
GTF2F1	HepG2	ENCFF002HYO	ENCSR265ZIS	2	4,472	yes
HLTF	HepG2	ENCFF048FDF	ENCSR647HOX	2	3,645	yes
HNRNPC	K562	ENCFF167CDB	ENCSR249ROI	2	516	yes
HNRNPL	K562	ENCFF917CBK	ENCSR795CAI	2	5,840	yes
HNRNPM	HepG2	ENCFF752JNY	ENCSR267UCX	2	13,423	yes
HNRNPUL1	HepG2	ENCFF610JVZ	ENCSR755TJC	2	467	yes
IGF2BP1	K562	ENCFF650LMV	ENCSR975KIR	2	5,706	yes
KHSRP	HepG2	ENCFF771QAU	ENCSR366DGX	2	5,289	yes
KHSRP	K562	ENCFF031FMO	ENCSR438GZQ	2	23,190	yes
LARP4	HepG2	ENCFF534JCV	ENCSR805SRN	2	5,429	yes
LARP4	K562	ENCFF947CIN	ENCSR888YTT	2	430	yes
LIN28B	K562	ENCFF061XNA	ENCSR970NKP	2	6,916	yes
LSM11	K562	ENCFF858EHN	ENCSR022BVV	2	1,339	yes
MATR3	HepG2	ENCFF587KKM	ENCSR290VLT	2	6,991	yes
MATR3	K562	ENCFF246EPM	ENCSR440SUX	2	4,866	yes
MBNL1	K562	ENCFF603WDI	ENCSR497XZI	2	1,202	yes
METTL1	K562	ENCFF125IDN	ENCSR498DQS	2	481	yes
MORC2	K562	ENCFF605AFN	ENCSR168VUE	2	891	yes
MTPAP	K562	ENCFF883KCN	ENCSR200DKE	2	1,083	yes
NCBP2	HepG2	ENCFF692RZM	ENCSR018RVZ	2	3,226	yes
NCBP2	K562	ENCFF886MLH	ENCSR484LTQ	2	406	no
NKRF	HepG2	ENCFF045HSV	ENCSR277DEO	2	3,794	yes
NOLC1	K562	ENCFF327YTD	ENCSR001VAC	2	1,570	yes
NONO	K562	ENCFF730QRI	ENCSR861PAR	2	11,333	yes
PCBP1	HepG2	ENCFF098REE	ENCSR256CHX	2	940	yes
PCBP2	HepG2	ENCFF642GNE	ENCSR339FUY	2	8,526	yes
PHF6	K562	ENCFF588ITW	ENCSR001KKZ	2	695	yes
PPIL4	K562	ENCFF559HGK	ENCSR197INS	2	659	yes
PRPF4	HepG2	ENCFF227EJF	ENCSR977OXG	2	9,159	yes
PTBP1	HepG2	ENCFF726SQU	ENCSR384KAN	2	7,988	yes
PTBP1	K562	ENCFF907HNN	ENCSR981WKN	2	7,608	yes
PUM1	K562	ENCFF094MQV	ENCSR308YNT	2	2,658	yes
QKI	HepG2	ENCFF704OCI	ENCSR570WLM	2	9,990	yes
QKI	K562	ENCFF786UOW	ENCSR366YOG	2	2,189	yes
RBFOX2	K562	ENCFF206RIM	ENCSR756CKJ	2	3,519	yes
RBM15	HepG2	ENCFF054VLU	ENCSR754NDA	2	1,500	yes
RPS3	HepG2	ENCFF301IJW	ENCSR766FAC	2	5,658	yes

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Table S1, continued

protein	cell line	ENCFF file	ENCSR experiment	reps	pairs	3 arms
RPS3	K562	ENCFF530HTL	ENCSR120EAR	2	2,372	yes
SAFB2	K562	ENCFF594EDV	ENCSR943MHU	2	3,744	yes
SLTM	K562	ENCFF696QWE	ENCSR000SSH	2	3,640	yes
SMNDC1	K562	ENCFF943XOU	ENCSR658IQB	2	2,465	yes
SND1	K562	ENCFF211TRO	ENCSR128VXC	2	4,132	yes
SRSF1	HepG2	ENCFF934ANS	ENCSR989VIY	2	1,791	yes
SRSF1	K562	ENCFF223KVR	ENCSR321PWZ	2	3,864	yes
STAU2	HepG2	ENCFF678FCX	ENCSR979EWD	2	848	yes
SUB1	HepG2	ENCFF335JTC	ENCSR406OOZ	2	3,297	yes
TAF15	HepG2	ENCFF566EAE	ENCSR841EQA	2	768	yes
TIA1	HepG2	ENCFF759KCD	ENCSR623VEQ	2	1,729	yes
TIA1	K562	ENCFF918KMT	ENCSR057DWB	2	5,883	yes
TRA2A	HepG2	ENCFF766OCH	ENCSR314UMJ	2	921	yes
U2AF1	K562	ENCFF640IHY	ENCSR862QCH	2	1,230	yes
U2AF2	HepG2	ENCFF721PWF	ENCSR202BFN	2	10,702	yes
UCHL5	HepG2	ENCFF573DXD	ENCSR490IEE	2	2,233	yes
XPO5	HepG2	ENCFF327VCE	ENCSR921SXC	2	4,615	yes
YBX3	HepG2	ENCFF185OEI	ENCSR735HOK	2	2,042	yes
YBX3	K562	ENCFF746XLE	ENCSR529FKI	2	17,979	yes
YWHAG	K562	ENCFF932UXC	ENCSR867ZVK	2	798	yes
ZNF622	K562	ENCFF285IGP	ENCSR657TZZ	2	14,969	yes
ZNF800	HepG2	ENCFF343TCO	ENCSR685AUR	2	2,938	yes
ZNF800	K562	ENCFF105HMJ	ENCSR586DGV	2	2,436	yes